

Ahmedabad University

Milestone 2(M2): Technical Foundation

Breast Cancer Survival Prediction

CSE516: Probabilistic Graphical Models

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Abstract

Breast cancer survival prediction is a challenging task because patient outcomes may be associated with information from multiple data modalities, including genomic measurements and histology-derived features. This project investigates the use of a Probabilistic Graphical Model (PGM) to represent and integrate these two modalities for survival prediction.

The project is based on a published multimodal probabilistic canonical correlation analysis framework for breast cancer survival prediction using The Cancer Genome Atlas (TCGA-BRCA) dataset. The graphical model represents genomics and histology-derived features as conditionally independent observations given a shared latent variable. During Milestone 2 (M2), the technical foundation was developed by implementing Gaussian probabilistic canonical correlation analysis, parameter estimation using maximum likelihood and the Expectation-Maximization (EM) algorithm, and posterior inference over the latent variable. A Cox proportional hazards model was used to evaluate the relationship between the learned representations and patient survival outcomes.

The experimental pipeline uses fixed five-fold data splits, training-set-based standardization, validation-based latent dimension selection, and Harrell's concordance index (C-index) for evaluation. Initial experiments were conducted using both the paper's PCA-based input configuration and a reduced input configuration containing genomic principal components and raw CellProfiler histology features. The reproduced paper baseline was also compared with the implemented methods.

The initial results show that predictive performance varies across input configurations and modeling approaches. For the PCA-based input, the EM-based multimodal model achieved a mean C-index of 63.44%, while the non-PGM baseline achieved 66.08%. For the reduced input configuration, the corresponding mean C-indices were 67.58% and 69.78%, respectively. These results establish an initial experimental baseline but do not demonstrate that the proposed PGM outperforms the non-PGM alternative. The experiments also revealed differences in EM convergence behavior across the two input configurations.

Overall, M2 establishes the core graphical model, parameter-learning procedures, inference methods, and evaluation pipeline required for further investigation. The next milestone will focus on improving and extending the experimental analysis, examining model limitations, and evaluating potential improvements to the framework.

1 Introduction and Problem Framing

1.1 Background and Motivation

Breast cancer is a major healthcare concern, and predicting patient survival is an important problem in medical data analysis. Patients may have different outcomes even when their disease characteristics appear similar. Developing computational methods that can learn meaningful patterns from patient data may help investigate the factors associated with these differences.

Modern biomedical datasets contain multiple types of information about the same patient. Genomic data describe molecular characteristics through measurements such as gene expression, while histology-derived features summarize characteristics extracted from tissue images. These modalities provide different views of the disease and may contain complementary information relevant to survival prediction.

However, combining multiple data modalities is not straightforward. The input features may have different statistical properties, dimensionalities, and relationships with the outcome. A model must therefore represent the relationships between the modalities while learning a useful representation of the underlying patient characteristics.

This project investigates the use of a Probabilistic Graphical Model (PGM) to integrate genomic and histology-derived features for breast cancer survival prediction. The project follows a published multimodal probabilistic canonical correlation analysis framework and develops an implementation for parameter learning, latent-variable inference, and survival prediction.

1.2 Problem Statement

The problem addressed in this project is to investigate whether a shared latent representation learned from genomic and histology-derived patient features can be used for breast cancer survival prediction.

Let X denote the genomic feature vector, Y denote the histology-derived feature vector, and Z denote a shared latent variable representing information common to the two modalities. The objective is to model the relationships among these variables and use the learned latent representation in a survival prediction pipeline.

The graphical model assumes that the two observed modalities are conditionally independent given the shared latent variable. This assumption is expressed as

$$X \perp Y \mid Z.$$

The corresponding joint probability distribution is factorized as

$$P(X, Y, Z) = P(Z)P(X \mid Z)P(Y \mid Z).$$

Here, $P(Z)$ represents the prior distribution of the latent variable, while $P(X \mid Z)$ and $P(Y \mid Z)$ describe how the genomic and histology-derived observations are generated from the shared representation.

The survival prediction component uses the learned representation in conjunction with a Cox proportional hazards model. Model performance is evaluated using Harrell's concordance index (C-index), which assesses how well predicted risk scores rank patients according to their observed survival outcomes.

1.3 Project Objectives

The main objectives of the project are:

1. **Develop the graphical model:** Implement the multimodal probabilistic model connecting genomic features, histology-derived features, and a shared latent variable.
2. **Implement parameter learning:** Develop maximum-likelihood estimation and the Expectation-Maximization (EM) algorithm for estimating model parameters.
3. **Perform latent-variable inference:** Estimate the latent representation using the joint observations and investigate inference using individual modalities.
4. **Build the survival prediction pipeline:** Connect the learned representations to a Cox proportional hazards model and evaluate survival prediction performance.
5. **Reproduce and compare baseline results:** Evaluate the implemented methods against the reproduced reference-paper baseline and a non-PGM baseline.
6. **Evaluate the technical foundation:** Examine predictive performance, EM convergence, and the effect of different input configurations and latent dimensions.

1.4 Scope of the M2 Milestone

The scope of Milestone 2 is to establish a working technical foundation for the project. This includes implementing the graphical model, developing parameter-learning and inference procedures, preparing the experimental pipeline, and obtaining initial quantitative results.

The experiments use the TCGA-BRCA dataset and fixed five-fold data splits. Two input configurations are considered: a PCA-based configuration following the reference paper and a reduced configuration combining genomic principal components with raw CellProfiler histology-derived features.

The evaluation focuses on the mean and standard deviation of the C-index across the five folds, parameter-learning behavior, and the influence of latent dimensionality on predictive performance. The reproduced reference-paper baseline provides a point of comparison for the implemented methods.

M2 is primarily concerned with implementation and initial evaluation rather than establishing a clinically deployable prediction system. The current results are therefore treated as experimental findings that require further investigation. Improvements to model behavior, additional experiments, and extensions to the framework remain part of the project's subsequent development.

1.5 Research Questions

The project is guided by the following research questions:

1. Can a multimodal probabilistic graphical model be implemented to represent the relationship between genomic and histology-derived features through a shared latent variable?
2. How do maximum-likelihood estimation and EM-based parameter learning compare in terms of predictive performance and convergence behavior?
3. How does the choice of input configuration and latent dimensionality affect survival prediction performance?
4. How does the implemented model compare with the reproduced reference-paper results and a non-PGM baseline?

2 Research Background and Reference-Paper Positioning

2.1 Reference Research

This project is based on the published research on multimodal probabilistic graphical modeling for breast cancer survival prediction. The reference framework combines genomic and histology-derived information using a probabilistic canonical correlation analysis approach. The objective is to learn a shared representation of the two modalities and investigate its usefulness for predicting patient survival.

The motivation for using a probabilistic model is that it provides an explicit representation of the relationships between observed variables and latent variables. Rather than treating the two modalities as unrelated collections of features, the model represents their relationship through a shared latent variable. This provides a probabilistic foundation for learning representations and performing inference.

The reference paper also provides a baseline for evaluating the implemented methods. Reproducing its reported results is important because it provides a point of comparison for the technical implementation and helps identify differences between the reference framework and the experiments conducted in this project.

2.2 Core Modeling Approach

The underlying graphical model contains three principal variables: genomic features X, histology-derived features Y, and a shared latent variable Z. The model structure is represented by

$$X \leftarrow Z \rightarrow Y.$$

The latent variable Z captures shared information used to model both observed modalities. Under the model's conditional-independence assumption, the joint distribution is factorized as

$$P(X, Y, Z) = P(Z)P(X \mid Z)P(Y \mid Z).$$

The model is based on Gaussian probabilistic canonical correlation analysis. Its parameters can be estimated using maximum-likelihood methods or the Expectation-Maximization (EM) algorithm. Once the model has been fitted, posterior inference can be performed to estimate the latent representation from both modalities jointly or from either modality individually.

The resulting latent representation is then used in the survival prediction pipeline. A Cox proportional hazards model connects the learned representation to patient survival outcomes, and Harrell's concordance index is used to evaluate predictive performance.

2.3 What Is Reproduced from the Reference Work?

The M2 implementation includes a paper-baseline reproduction using the authors' stored trained projection matrices, stored outputs, and survival-analysis code. The repository's saved results report agreement with the authors' stored per-fold C-index values, with a maximum difference of 0.00 percentage points for the reproduced comparison.

This reproduction is an important validation step, but its scope must be stated precisely. The original MATLAB training procedure was not rerun as part of this reproduction. Instead, the available trained artifacts and outputs were used to reproduce the reported evaluation. Consequently, the agreement validates the reproduced evaluation against the available author-provided results, rather than independently verifying the complete original training process.

The project also implements its own experimental pipeline for fitting probabilistic CCA models, estimating parameters, performing latent-variable inference, and evaluating survival prediction under different input configurations. These experiments provide additional results beyond the stored paper-baseline evaluation.

2.4 Implementation Work in M2

The principal technical work completed during M2 consists of the following:

- Implementation of Gaussian probabilistic canonical correlation analysis with maximum-likelihood parameter estimation and EM-based learning.
- Development of posterior inference for the shared latent variable using joint genomic and histology-derived observations, as well as modality-specific observations.
- Integration of learned representations with a Cox proportional hazards model for survival prediction.
- Construction of an experimental pipeline using fixed five-fold data splits, training-set-based standardization, and validation-based latent dimension selection.
- Evaluation of predictive performance using Harrell's C-index, alongside analysis of EM convergence and latent dimensionality.

- Generation of saved experimental results, comparison tables, and visualizations to support the M2 evaluation.

These components establish a reproducible technical foundation for further experiments. They should be understood as implementation and evaluation work based on an existing research framework, rather than evidence that an entirely new graphical model has been proposed.

2.5 Positioning of the Current Results

The initial results show that the relative performance of the implemented methods depends on the input configuration and learning procedure. In the PCA-based configuration, the EM-based multimodal model achieves a mean C-index of 63.44%, compared with 66.08% for the non-PGM baseline. In the reduced input configuration, the corresponding values are 67.58% and 69.78%.

These results do not establish that the PGM consistently outperforms the non-PGM baseline. Instead, they provide an initial assessment of the implemented framework and identify areas that require further investigation, including convergence behavior, representation learning, and predictive performance.

The contribution of M2 is therefore the establishment of the modeling, inference, and evaluation pipeline, together with initial quantitative comparisons. Further claims about improvements over the reference method or the value of additional extensions will require supporting experiments in subsequent milestones.

3 Dataset and Preprocessing

3.1 Dataset Description

The project uses the TCGA-BRCA dataset for breast cancer survival prediction. The dataset contains patient-level information from two modalities: genomic measurements and histology-derived features. These modalities provide different representations of patient characteristics and are combined through the shared latent variable in the probabilistic graphical model.

The dataset documentation in the project repository reports 974 patients. The experimental pipeline uses fixed five-fold data splits, with each fold containing training, validation, and test partitions. The documented partition sizes are 585 training patients, 146 validation patients, and 243 test patients. The dataset documentation also reports 95 deaths.

Genomic data provide molecular measurements, whereas histology-derived features summarize characteristics extracted from tissue images using computational feature extraction. The model operates on these numerical feature representations rather than directly processing raw histology images.

3.2 Input Modalities

Two input modalities are used in the graphical model:

- **Genomic modality (X):** Represents genomic measurements through numerical features derived from the dataset.
- **Histology modality (Y):** Represents tissue characteristics through numerical features extracted using CellProfiler.

The two modalities are used as observations of a shared latent variable. The feature representations supplied to the model depend on the experimental configuration.

3.3 Experimental Input Configurations

Two input configurations are evaluated during M2. They allow the implemented methods to be examined under the paper-based representation and a reduced alternative.

Table 1: Input configurations used in M2 experiments

Configuration	**Genomic Input**	**Histology Input**
Paper PCA configuration	50 genomic principal components	50 histology principal components
Reduced configuration	50 genomic principal components	215 raw CellProfiler histology features

In the paper PCA configuration, both modalities are represented using 50 principal components. This configuration follows the input representation used for comparison with the reference work.

In the reduced configuration, the genomic input consists of 50 principal components, while the histology input contains 215 raw CellProfiler features. These histology features are not principal components. Keeping this distinction explicit is necessary when interpreting differences between the experimental results.

The repository also contains a higher-dimensional feature representation with 1,000 genomic features and 1,075 histology features. Fitting the full representation with the current model was deferred to M3 because of covariance-estimation and sample-size constraints.

3.4 Data Splitting Strategy

The experimental pipeline uses five fixed data splits. Each split contains separate training, validation, and test partitions. The documented partition sizes are summarized in Table 2.

Table 2: Documented patient partition sizes

Partition	**Number of Patients**
Training	585
Validation	146
Test	243
Total	974

The training partition is used to estimate model parameters. The validation partition is used for selecting the latent dimensionality, while the test partition is reserved for evaluating predictive performance.

Using fixed splits supports consistent comparisons between the implemented methods. Keeping the test data separate from model selection also reduces the risk of using test performance to make modeling decisions.

3.5 Preprocessing and Standardization

Feature standardization is performed using statistics estimated from the training partition. The resulting training-derived transformation is then applied to the corresponding validation and test partitions.

This approach is important because calculating preprocessing statistics using all partitions would allow information from validation or test observations to influence the training process. Fitting the standardization procedure on training data only helps prevent this form of data leakage.

The preprocessing strategy is applied within the experimental pipeline so that model fitting and evaluation follow a consistent procedure across the fixed data splits.

3.6 Dataset Limitations

The dataset contains substantially more features than patients in some of its input representations. This creates challenges for estimating covariance matrices and fitting Gaussian latent-variable models reliably. The project therefore evaluates reduced-dimensional representations during M2 rather than fitting the full feature representation with the current model.

In addition, the reported number of deaths is smaller than the total number of patients. Survival prediction performance must consequently be interpreted with care, particularly when comparing methods across a limited number of folds.

The current dataset and preprocessing pipeline provide a foundation for evaluating multimodal probabilistic modeling. Further work will be required to assess the behavior of the framework with the full feature representation and to determine whether its performance remains consistent under additional experimental settings.

4 PGM Methodology and Technical Foundation

4.1 Overview of the Proposed Framework

The project implements a multimodal probabilistic modeling framework for breast cancer survival prediction. The framework combines genomic and histology-derived features through a shared latent variable and uses the learned representation for downstream survival analysis.

Let X represent the genomic feature vector, Y represent the histology-derived feature vector, and Z represent the shared latent variable. The observed variables contain information about

a patient, while the latent variable provides a lower-dimensional representation through which the two modalities are modeled.

The framework consists of three main stages:

1. Learning the probabilistic relationship between the two modalities and the shared latent variable.
2. Performing inference to estimate the latent representation from the available observations.
3. Using the learned representation in a Cox proportional hazards model to predict relative survival risk.

4.2 Graphical Model Structure

The graphical model used in this project is represented by the following structure:

$$X \leftarrow Z \rightarrow Y$$

Here, Z is the common parent of the observed variables X and Y. The model assumes that the two modalities are conditionally independent given Z.

This conditional-independence assumption is written as

$$X \perp Y \mid Z.$$

Under this assumption, the joint probability distribution factorizes as

$$P(X, Y, Z) = P(Z)P(X \mid Z)P(Y \mid Z).$$

The factorization expresses the model's structure: the latent variable has a prior distribution, and each observed modality is modeled conditionally on that latent variable.

The graphical structure is specified in advance in the current implementation. The project estimates model parameters within this structure rather than learning the graph structure itself.

4.3 Gaussian Probabilistic Canonical Correlation Analysis

The project uses Gaussian probabilistic canonical correlation analysis (PCCA) to represent the relationship between the two observed modalities through a shared latent variable.

A general linear-Gaussian formulation is

$$Z \sim \mathcal{N}(0, I_d),$$

$$X = W_X Z + \mu_X + \epsilon_X,$$

$$Y = W_Y Z + \mu_Y + \epsilon_Y,$$

where d is the latent dimensionality, W_X and W_Y are modality-specific loading matrices, and μ_X and μ_Y are modality-specific mean vectors.

The residual terms are modeled as Gaussian noise:

$$\epsilon_X \sim \mathcal{N}(0, \Psi_X), \qquad \epsilon_Y \sim \mathcal{N}(0, \Psi_Y).$$

In this formulation, I_d is the d-dimensional identity matrix, while Ψ_X and Ψ_Y represent the noise covariance matrices. The shared latent variable allows the model to represent statistical relationships between the two modalities.

The precise parameter constraints and covariance assumptions are determined by the implemented model configuration. The equations above describe the general linear-Gaussian formulation underlying the approach.

4.4 Parameter Learning

Parameter learning estimates the model parameters that best explain the observed training data. The project implements two approaches: maximum-likelihood estimation and the Expectation-Maximization algorithm.

4.4.1 Maximum-Likelihood Estimation

Maximum-likelihood estimation (MLE) selects parameters that maximize the likelihood of the observed training data.

Let θ denote the collection of model parameters. The maximum-likelihood objective is

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \log P(X, Y \mid \theta).$$

Because the latent variable Z is not directly observed, estimating the marginal likelihood requires accounting for its possible values:

$$P(X, Y \mid \theta) = \int P(X, Y, Z \mid \theta)\, dZ.$$

The MLE implementation provides one approach to fitting the probabilistic model. Its estimated parameters are subsequently used for latent-variable inference and downstream survival prediction.

4.4.2 Expectation-Maximization

The Expectation-Maximization (EM) algorithm provides an iterative approach to parameter estimation when the model includes latent variables.

EM alternates between two steps:

- **Expectation step (E-step):** Estimate the posterior distribution of the latent variable under the current parameter estimates and calculate the expected complete-data log-likelihood.
- **Maximization step (M-step):** Update the model parameters to maximize the expected complete-data log-likelihood obtained in the E-step.

The EM objective can be expressed as

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{Z|X,Y,\theta^{(t)}} \left[\log P(X, Y, Z \mid \theta)\right],$$

where $\theta^{(t)}$ denotes the parameter estimates at iteration t.

The parameter update is then

$$\theta^{(t+1)} = \arg\max_{\theta} Q(\theta \mid \theta^{(t)}).$$

These steps are repeated until the configured convergence criterion is satisfied or the maximum iteration limit is reached. The project records convergence behavior as part of the experimental evaluation.

4.5 Latent-Variable Inference

After the model has been fitted, posterior inference estimates the latent representation from the observed features. The project considers inference using both modalities together and using either modality individually.

4.5.1 Joint-Modality Inference

When genomic and histology-derived features are both available, the posterior distribution is

$$P(Z \mid X, Y).$$

This posterior combines information from the two modalities and provides the joint latent representation used in the multimodal prediction pipeline.

4.5.2 Genomics-Only Inference

When only genomic features are considered, inference uses

$$P(Z \mid X).$$

This provides a modality-specific representation based on genomic observations.

4.5.3 Histology-Only Inference

When only histology-derived features are considered, inference uses

$$P(Z \mid Y).$$

This provides a modality-specific representation based on histology observations.

For each inference setting, a point estimate of the latent variable can be obtained from the corresponding posterior distribution. Where the implementation uses the maximum a posteriori (MAP) estimate, it is defined as

$$\hat{Z}_{\text{MAP}} = \arg\max_{Z} P(Z \mid \text{observations}).$$

The available observations in this expression depend on the inference setting. These modality-specific calculations allow the project to evaluate the learned representation using joint or individual modalities.

4.6 Survival Prediction

The inferred latent representation is connected to a Cox proportional hazards model to evaluate its usefulness for survival prediction.

For a patient with latent representation Z, the Cox model expresses the hazard function as

$$h(t \mid Z) = h_0(t) \exp(\beta^\top Z),$$

where $h(t \mid Z)$ is the hazard at time t, $h_0(t)$ is the baseline hazard function, and β is the vector of regression coefficients.

The term $\beta^\top Z$ defines the patient's relative log-risk under the model. The corresponding risk score is

$$r(Z) = \beta^\top Z.$$

The Cox model is fitted using the available survival outcomes and the representation supplied by the selected inference method. Model performance is then evaluated using Harrell's concordance index.

4.7 Posterior Differential Entropy

The project also includes analysis of posterior uncertainty through differential entropy. For a continuous latent variable with posterior density $p(z \mid \mathcal{O})$, where $\mathcal{O}$ denotes the available observations, differential entropy is defined as

$$H(Z \mid \mathcal{O}) = -\int p(z \mid \mathcal{O}) \log p(z \mid \mathcal{O})\, dz.$$

Posterior entropy summarizes the spread of the posterior distribution. For a Gaussian posterior with covariance matrix Σ in d dimensions, the differential entropy is

$$H(Z \mid \mathcal{O}) = \frac{1}{2} \log \left((2\pi e)^d \det \Sigma \right).$$

This analysis provides an additional way to examine the uncertainty associated with the inferred latent representation. Differential entropy is not itself a measure of survival prediction accuracy; it describes uncertainty in the posterior distribution.

4.8 Methodological Summary

The M2 methodology establishes a complete modeling path from multimodal feature inputs to survival prediction. The graphical model defines the relationship between the modalities, PCCA provides the linear-Gaussian latent-variable formulation, MLE and EM provide parameter-learning approaches, and posterior inference estimates the latent representation.

The inferred representation is subsequently evaluated through a Cox proportional hazards model. Together, these components form the technical foundation for the initial experiments and comparisons reported in the following sections.

5 Implementation and Experimental Pipeline

5.1 Implementation Overview

The M2 implementation is organized into separate modules for probabilistic modeling, data preparation, survival analysis, experiment execution, and result visualization. This modular organization allows the individual components to be developed and evaluated within a common experimental pipeline.

The implementation is maintained in the project's GitHub repository under the `M2_G02/Code/` directory. The source code is organized into a main execution script, reusable modeling modules, experiment scripts, and utilities for generating summaries and figures.

The principal implementation files are listed in Table 3.

5.2 Data Loading and Preparation

The experimental pipeline loads the prepared feature representations and survival outcomes from the project's data files. The inputs are organized according to the experimental configuration and fixed data splits.

The pipeline uses the paper PCA configuration and the reduced input configuration described in Section 5. Each configuration is evaluated using the corresponding genomic and histology-derived features.

Standardization is fitted using the training partition and then applied to the validation and test partitions. This ensures that the preprocessing transformation does not use statistics

Table 3: Principal M2 implementation modules

File	Purpose
`src/pcca.py`	Implements the probabilistic CCA modeling and parameter-learning functionality.
`src/data.py`	Handles data loading and preparation for the experiments.
`src/survival.py`	Provides functionality for survival analysis and prediction evaluation.
`src/run_experiments.py`	Runs the principal experimental comparisons.
`src/sweep_latent_dim.py`	Evaluates different latent dimensionalities.
`src/make_figures.py`	Generates visualizations from the experimental results.
`src/write_summary.py`	Produces a summary of the experimental results.
`src/paper_baseline.py`	Supports the reproduction and evaluation of the reference-paper baseline.
`run_all.py` and `run_all.sh`	Provide entry points for running the experiment workflow.

calculated from the held-out partitions.

5.3 Model Fitting

The modeling stage fits the probabilistic CCA model using the selected parameter-learning method. The M2 experiments include maximum-likelihood estimation and EM-based parameter learning.

For each experiment, the selected input representation and latent dimensionality determine the model configuration. The fitted model is subsequently used to infer latent representations for the relevant data partitions.

The project also evaluates joint-modality and modality-specific representations. This allows survival prediction to be examined using both genomic and histology-derived information together, as well as using either modality individually.

5.4 Latent Dimension Selection

The latent dimensionality determines the number of dimensions in the shared representation. Since the most appropriate dimensionality is not known in advance, the implementation evaluates a predefined set of candidate values.

The latent dimension is selected using validation-set C-index. The test partition is reserved for evaluating the selected configuration rather than choosing among candidate dimensions.

The saved experimental results report selected dimensions of 30 for CCA-MLE and 50 for EM under the paper PCA configuration. For the reduced configuration, the selected dimensions

Table 4: Latent dimensionality candidates evaluated in M2

Input Configuration	**Method**	**Candidate Dimensions**
Paper PCA	CCA-MLE and EM	5, 10, 20, 30, 50
Reduced input	CCA-MLE and EM	2, 5, 10, 20, 30

are 20 for CCA-MLE and 10 for EM.

These values describe the configurations selected in the current experiments. They do not establish that the same dimensions will be optimal for other datasets or feature representations.

5.5 Survival Prediction and Evaluation

After fitting the probabilistic model and inferring the latent representation, the pipeline uses a Cox proportional hazards model for survival prediction.

Harrell's concordance index (C-index) is used to evaluate the agreement between predicted risk rankings and observed survival outcomes. Results are summarized across the five fixed folds using the mean and standard deviation.

The evaluation includes the multimodal model, genomics-only and histology-only settings, and a non-PGM baseline. The reproduced paper baseline is also included in the comparison where applicable.

5.6 Experiment Execution

The repository provides execution scripts for running the experimental workflow. The `run_all.sh` script is documented as an entry point for the main experiment pipeline, latent dimension sweeps, and figure generation.

The general workflow is summarized below:

1. Load the prepared feature data, survival outcomes, and fixed data splits.
2. Apply preprocessing using statistics derived from the training partition.
3. Fit the probabilistic model using the selected parameter-learning method.
4. Infer latent representations for the selected observation configuration.
5. Fit the survival prediction model and calculate the C-index.
6. Evaluate candidate latent dimensions using the validation partition.
7. Save experimental outputs and generate comparison tables and figures.

The workflow separates model fitting, validation-based configuration selection, and final test evaluation. This separation supports consistent comparisons across the methods and input configurations.

5.7 Saved Results and Visualization

The repository contains saved experimental outputs in the `M2_G02/Results/` directory. These include JSON result files, a KPI table in CSV format, a summary document, and figures illustrating the model structure and experimental findings.

The principal saved outputs include:

- `m2_paper_baseline.json`: results associated with the reproduced paper baseline.
- `m2_results_main.json`: results for the main reduced-input experiments.
- `m2_results_paperpca.json`: results for the paper PCA input configuration.
- `m2_latent_dim_sweep_main.json`: latent dimension sweep results for the reduced configuration.
- `m2_latent_dim_sweep_paperpca.json`: latent dimension sweep results for the paper PCA configuration.
- `m2_kpi_table.csv`: a tabular summary of key performance indicators.
- `SUMMARY.md`: a summary of the saved experimental findings.

The figure-generation module uses the saved results to produce visualizations for the graphical model, EM convergence, C-index comparisons, latent dimension sweeps, posterior entropy, and comparison with the reference paper.

These outputs provide the quantitative and visual evidence used in the results section of this report.

5.8 Implementation Status at M2

The repository contains the modeling modules, experiment scripts, and saved outputs needed to document the M2 technical foundation. The saved results provide evidence of experiments conducted and recorded in the project.

However, the presence of execution scripts does not, by itself, establish that the complete workflow has been rerun successfully in a fresh environment. Reproducibility also depends on the required data files, software dependencies, configuration settings, and execution environment.

The current M2 implementation therefore establishes the core experimental pipeline while leaving further validation, extension to the full feature representation, and additional experiments for subsequent development.

6 Experimental Setup and Evaluation Metrics

6.1 Experimental Objective

The experiments conducted during Milestone 2 evaluate the technical implementation of the multimodal probabilistic graphical model and examine its performance in breast cancer survival

prediction.

The evaluation focuses on four main aspects: predictive performance, comparison between parameter-learning approaches, sensitivity to latent dimensionality, and convergence behavior. The experiments also compare the implemented methods with a non-PGM baseline and the reproduced reference-paper results.

Two input configurations are evaluated: the paper PCA configuration and the reduced input configuration described in Section 5. The results are analyzed separately for these configurations because they use different feature representations.

6.2 Data Splits and Evaluation Protocol

The experiments use five fixed data splits, each containing training, validation, and test partitions. Model parameters are estimated using the training partition, and validation performance is used to select the latent dimensionality.

After selecting the configuration, predictive performance is evaluated on the corresponding held-out test partition. Results are summarized across the five folds using the mean and standard deviation.

This protocol is intended to support consistent comparisons between methods while keeping the test partition separate from model selection.

6.3 Methods Included in the Comparison

The evaluation includes the following methods and input settings.

Table 5: Methods and configurations evaluated during M2

Method or Setting	**Description**
CCA-MLE	Probabilistic CCA fitted using maximum-likelihood parameter estimation.
EM-based PGM	Probabilistic CCA fitted using the Expectation-Maximization algorithm.
Non-PGM baseline	A non-probabilistic-graphical-model baseline used for predictive comparison.
Paper baseline	Reproduced reference-paper results based on available author-provided trained artifacts and evaluation outputs.
Genomics-only	Survival prediction using a representation inferred from genomic observations.
Histology-only	Survival prediction using a representation inferred from histology-derived observations.
Multimodal	Survival prediction using the representation inferred jointly from genomic and histology-derived observations.

The paper baseline is distinguished from the newly fitted models because the reproduction uses available author-provided artifacts rather than rerunning the original MATLAB training procedure.

6.4 Primary Evaluation Metric: Harrell's C-index

Harrell's concordance index (C-index) is the primary metric used to evaluate survival prediction performance. It measures how consistently predicted risk scores rank patients whose observed survival outcomes can be compared.

A concordance index of 0.5 corresponds approximately to chance-level ranking, while higher values indicate better concordance between predicted risk rankings and observed survival outcomes.

The C-index can be expressed conceptually as

$$C = \frac{\text{Number of concordant comparable pairs} + \frac{1}{2}\text{Number of tied pairs}}{\text{Number of comparable pairs}}.$$

A pair is concordant when the predicted risk ranking agrees with the ordering of the observed survival outcomes for that comparable pair. The treatment of censored observations determines which pairs are considered comparable.

For each method and input configuration, the C-index is calculated for the test partition of each fold. The mean and standard deviation across the five folds are then reported.

The mean summarizes the average predictive performance, while the standard deviation indicates the variability across folds. These descriptive statistics are used to compare the initial results; they do not, by themselves, establish statistical significance.

6.5 Log-Likelihood Evaluation

The experimental outputs also record training and test log-likelihood per sample for the probabilistic models. Log-likelihood measures how well a fitted probabilistic model assigns probability density to the observed data under its specified distribution.

For observed data D and model parameters θ, the log-likelihood is

$$\log P(D \mid \theta).$$

The reported values provide an additional perspective on model fitting and generalization beyond survival prediction performance.

However, log-likelihood values should only be compared directly when the evaluated models use compatible likelihood definitions and comparable observed variables. In particular, the likelihood values obtained from the paper PCA and reduced input configurations are not directly comparable because the feature representations differ.

6.6 Computation Time

The saved experimental results also record model-fitting time. Computation time provides a practical measure of the cost associated with fitting the model under a particular input configuration and parameter-learning approach.

The recorded time depends on factors such as the number of features, latent dimensionality, convergence behavior, and execution environment. It should therefore be interpreted as an observed implementation result rather than a hardware-independent measure of algorithmic efficiency.

6.7 EM Convergence Evaluation

The convergence behavior of the EM algorithm is evaluated by examining the saved iteration counts and convergence outcomes for the five folds.

Two aspects are considered:

- **Convergence status:** Whether the algorithm satisfies its configured convergence criterion within the allowed number of iterations.
- **Iteration behavior:** How many iterations are required and whether the optimization exhibits stable progress or reaches the iteration limit.

The maximum iteration limit in the saved experiments is 300. The recorded results show that all five folds converged for the paper PCA configuration, while only two of the five folds converged for the reduced input configuration. The other three reduced-input runs reached

7 Results and Discussion

7.1 Overview of Experimental Results

This section presents the initial experimental results obtained from the M2 implementation of the multimodal probabilistic graphical model. The evaluation covers predictive performance, comparison with baseline methods, latent dimensionality, EM convergence, and posterior uncertainty.

The results are reported separately for the paper PCA configuration and the reduced input configuration. The two configurations use different feature representations, so their results are interpreted independently rather than as a direct controlled comparison.

The figures in this section are the saved visualizations in the project's `M2_G02/Results/` directory. They provide graphical summaries of the implemented model and its experimental evaluation.

7.2 Graphical Model Structure

Figure 1 shows the graphical model used in the project. The genomic and histology-derived observations share a common latent variable, which provides the basis for multimodal representation learning.

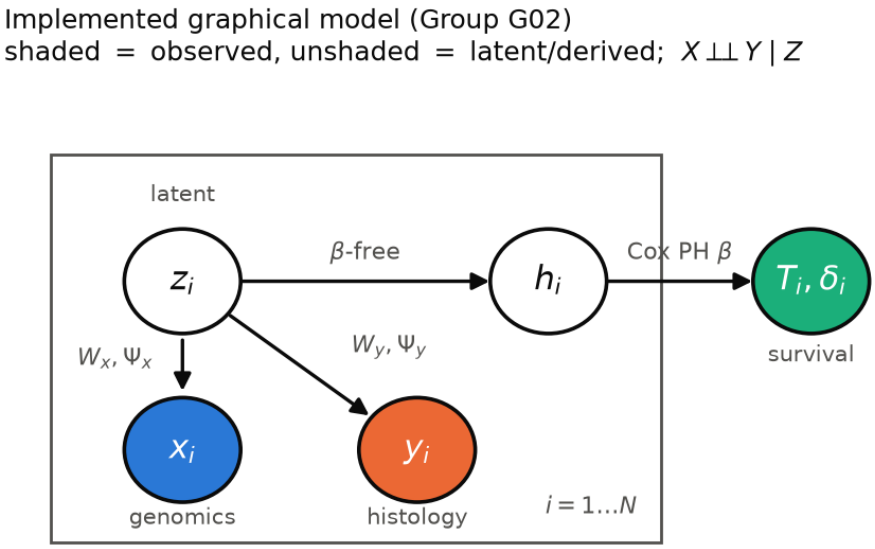


Figure 1: Graphical structure of the multimodal probabilistic model. The shared latent variable connects the genomic and histology-derived observations.

The graph represents the conditional-independence assumption $X \perp Y \mid Z$. The graph structure is specified in advance, while the model parameters are estimated during training. The figure therefore illustrates the model being fitted rather than a graph structure learned from the data.

7.3 Predictive Performance: Paper PCA Configuration

The paper PCA configuration uses 50 genomic principal components and 50 histology principal components. Table 6 reports the mean C-index and standard deviation across five folds for the implemented methods.

Table 6: Survival prediction results for the paper PCA configuration. Values are mean C-index and standard deviation across five folds, expressed as percentages.

Method	**Multimodal**	**Genomics**	**Histology**
CCA-MLE, $d = 30$	65.43 ± 6.58	67.31 ± 8.61	57.34 ± 6.46
EM, $d = 50$	63.44 ± 8.09	69.95 ± 8.68	56.93 ± 8.38
Non-PGM baseline	66.08 ± 6.97	67.82 ± 7.38	57.64 ± 8.49

For the multimodal setting, the CCA-MLE model achieves a mean C-index of 65.43%, while the EM-based model achieves 63.44%. The non-PGM baseline achieves 66.08%, which is higher than both implemented PGM results for this configuration.

The EM-based model achieves a higher mean C-index in the genomics-only setting than in the multimodal setting. This indicates that, for the selected configuration, adding the second modality does not automatically improve survival-risk ranking.

The histology-only results are lower than the genomics-only results across all three methods in this table. These observations describe the saved experimental results and do not establish that genomic features will always be more informative than histology-derived features in other settings.

Figure 2 visualizes the C-index comparisons for the paper PCA configuration.

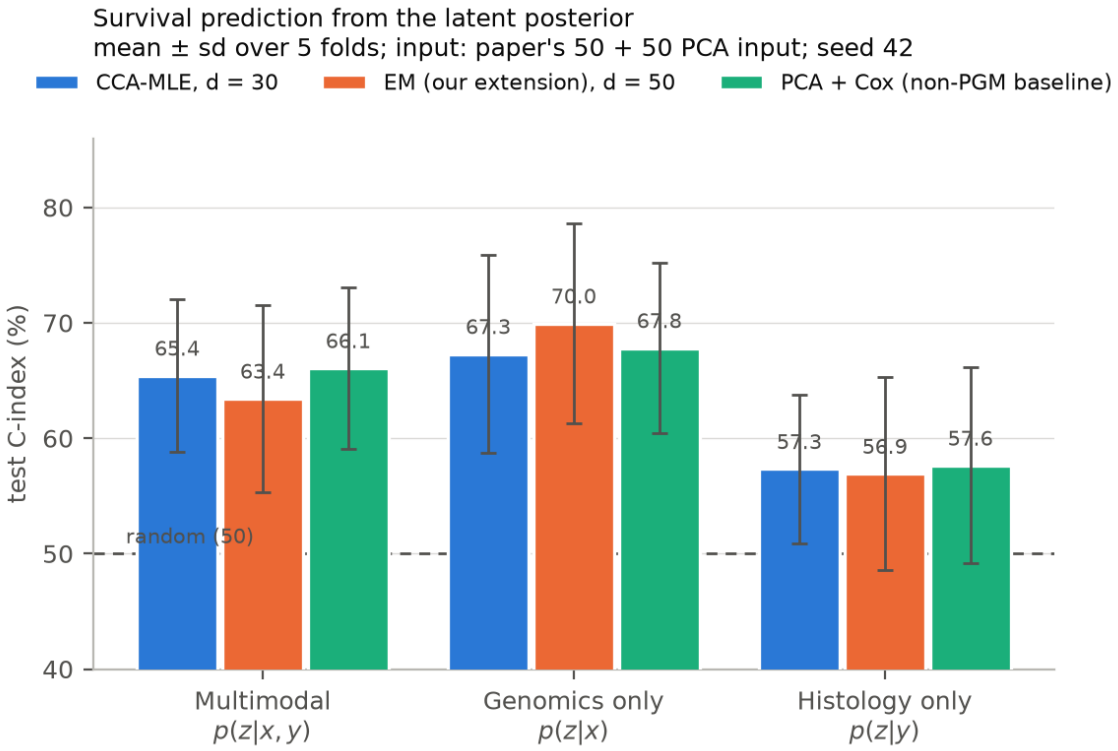


Figure 2: C-index comparison for the paper PCA input configuration.

7.4 Predictive Performance: Reduced Input Configuration

The reduced configuration uses 50 genomic principal components and 215 raw CellProfiler histology features. Unlike the paper PCA configuration, the histology features in this experiment are not principal components.

Table 7 presents the corresponding predictive performance.

Table 7: Survival prediction results for the reduced input configuration. Values are mean C-index and standard deviation across five folds, expressed as percentages.

Method	**Multimodal**	**Genomics**	**Histology**
CCA-MLE, $d = 20$	65.95 ± 10.55	67.26 ± 10.50	59.23 ± 3.91
EM, $d = 10$	67.58 ± 7.76	72.11 ± 6.17	60.30 ± 5.97
Non-PGM baseline	69.78 ± 6.32	67.82 ± 7.38	62.78 ± 8.45

For the reduced configuration, the EM-based model achieves a mean multimodal C-index of 67.58%, compared with 65.95% for CCA-MLE. However, the non-PGM baseline achieves 69.78%, exceeding both PGM results.

The EM-based model achieves the highest mean C-index among the listed methods in the genomics-only setting, at 72.11%. In the histology-only setting, the non-PGM baseline achieves the highest mean C-index, at 62.78%.

The fold-to-fold variability also differs between methods. For example, the CCA-MLE multimodal result has a standard deviation of 10.55 percentage points, whereas the non-PGM multimodal baseline has a standard deviation of 6.32 percentage points. These differences indicate that average performance alone does not describe the full behavior of the models.

Figure 3 provides the corresponding visual comparison for the main experiments.

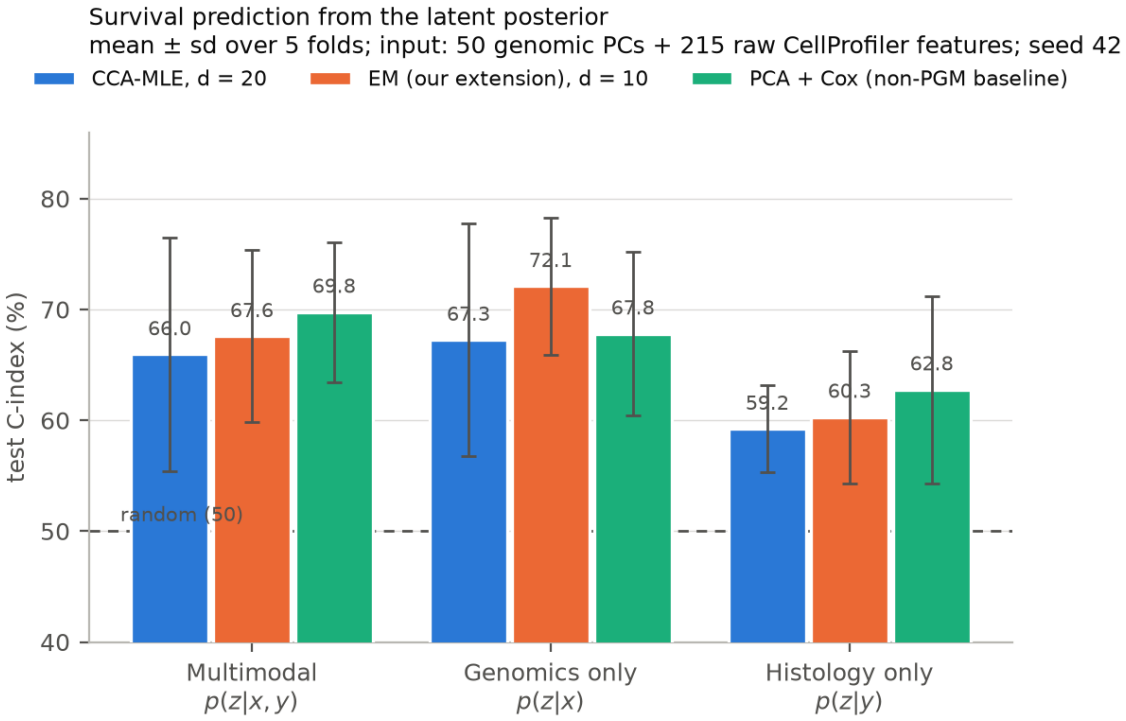


Figure 3: C-index comparison for the reduced input configuration.

7.5 Comparison with Reference-Paper Results

The reproduced paper baseline provides a reference point for evaluating the implementation. Table 8 summarizes the saved multimodal C-index results for the reference methods and the implemented methods using the paper PCA input configuration.

Among the listed reference methods, SCCA/OPD has the highest reported mean C-index of 68.98%. The implemented EM-based multimodal model has a lower mean C-index of 63.44%, while CCA-MLE achieves 65.43%.

The repository records a failed result for the reference GN-SCCA/HD configuration. Since no valid C-index is available for that configuration in the saved results, it is not included in the numerical comparison table.

Table 8: Multimodal C-index comparison with the reference-paper results. Values are mean and standard deviation in percentage points across five folds.

Method	**C-index (%)**
Reference PCA	66.47 ± 6.86
Reference SCCA/OPD	68.98 ± 6.16
Reference GN-SCCA/PD	65.47 ± 10.14
Reference GN-SCCA/OPD	66.85 ± 9.47
CCA-MLE, $d = 30$	65.43 ± 6.58
EM, $d = 50$	63.44 ± 8.09
Non-PGM baseline	66.08 ± 6.97

The paper-baseline reproduction uses author-provided trained projection matrices, stored outputs, and survival-analysis code. The repository reports agreement with the authors' stored per-fold C-index values to a maximum difference of 0.00 percentage points. This agreement validates the reproduced evaluation against those stored results, but it does not establish that the original MATLAB training procedure has been independently reproduced.

Figure 4 presents the saved visual comparison with the reference-paper results.

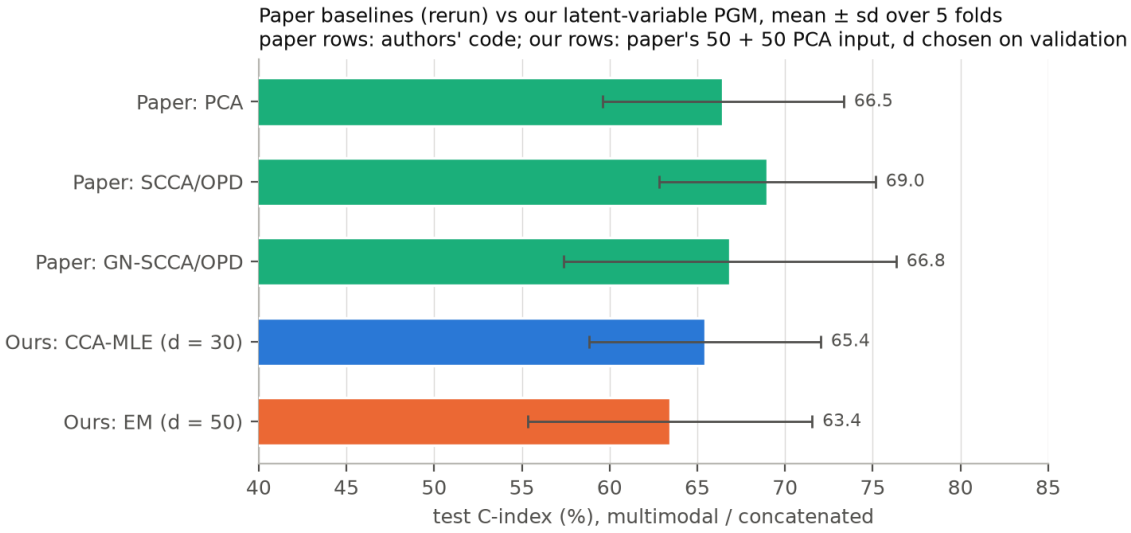


Figure 4: Comparison between the implemented methods and the reference-paper results.

7.6 Log-Likelihood and Computation Time

The saved results also include training and test log-likelihood per sample, together with model-fitting time. Table 9 summarizes these measurements.

For the PCA configuration, EM has a slightly higher mean training log-likelihood than CCA-MLE, but its mean test log-likelihood is lower. The saved results also show a longer fitting time for EM.

For the reduced configuration, EM has higher reported training and test log-likelihood values than CCA-MLE, but it also requires more fitting time. These values should not be interpreted

Table 9: Saved log-likelihood and fitting-time results

Configuration and Method	Train LL/sample	Test LL/sample	Time (s)
PCA, CCA-MLE	-132.16 ± 0.38	-157.10 ± 4.63	0.01
PCA, EM	-131.56 ± 0.37	-158.68 ± 5.19	0.14
Reduced, CCA-MLE	107.52 ± 0.79	93.02 ± 2.62	0.02
Reduced, EM	338.06 ± 1.88	259.20 ± 14.76	2.53

as proof that EM is universally better or worse because the behavior depends on the model configuration and input representation.

Importantly, the log-likelihood values from the PCA and reduced configurations are not directly comparable. The configurations use different observed feature representations, which changes the likelihood being evaluated. Log-likelihood should therefore be interpreted within each compatible configuration rather than used to rank the two configurations against each other.

7.7 EM Convergence Behavior

Convergence is an important part of evaluating the EM-based model because parameter estimates depend on the behavior of the iterative optimization procedure.

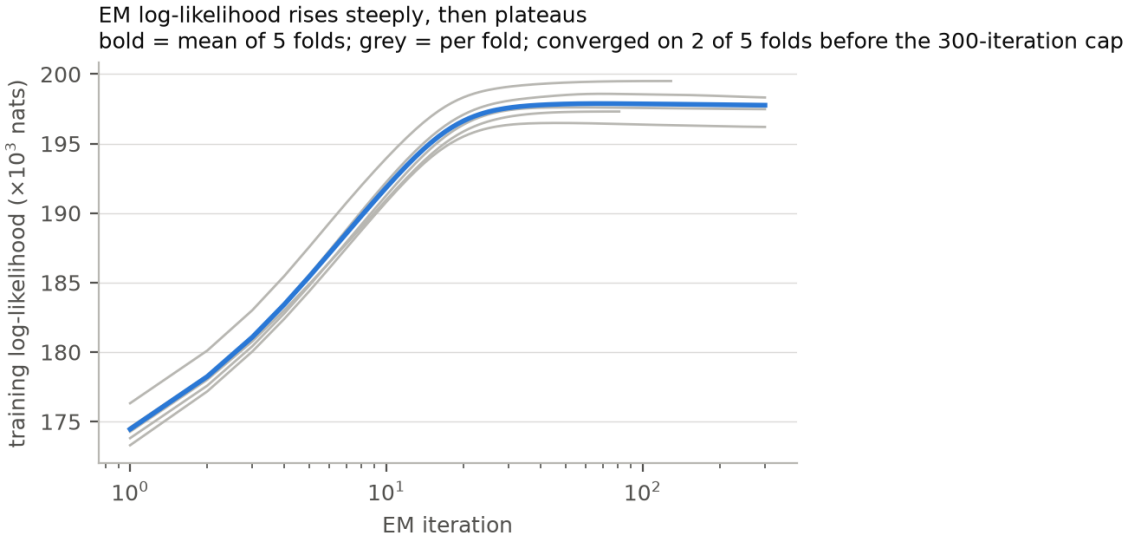


Figure 5: Saved EM convergence results for the evaluated input configurations.

For the paper PCA configuration, all five folds are recorded as converged, with convergence occurring between 75 and 104 iterations. In contrast, only two of the five folds converge for the reduced configuration. The other three runs reach the maximum limit of 300 iterations.

This difference indicates that the EM algorithm behaves differently under the two input configurations. The reduced configuration's convergence behavior deserves further investigation before drawing strong conclusions about the stability of its fitted parameters.

The results do not, by themselves, identify the cause of the difference. Potential explanations should be investigated through further diagnostics of parameter updates, likelihood traces,

initialization, and covariance estimation rather than assumed from the convergence plot alone.

7.8 Effect of Latent Dimensionality

The project evaluates multiple candidate latent dimensions to investigate how representation size affects survival prediction. Latent dimension selection is based on validation-set C-index.

For the paper PCA configuration, the candidates are 5, 10, 20, 30, and 50. For the reduced configuration, the candidates are 2, 5, 10, 20, and 30.

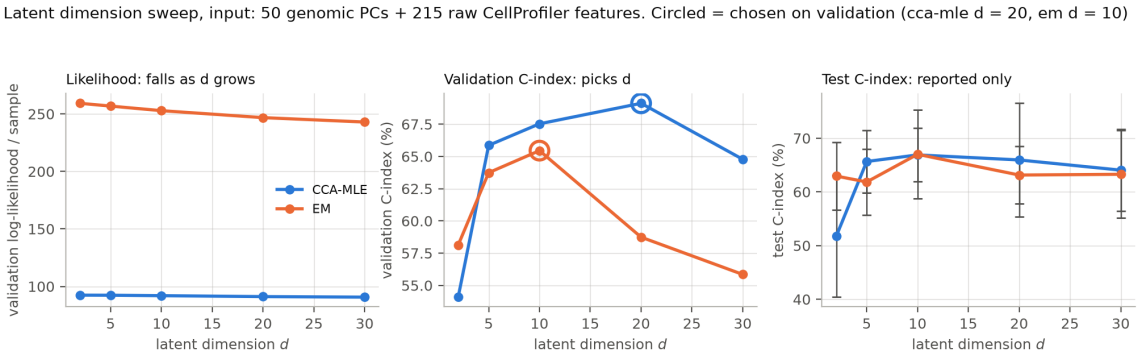


Figure 6: Latent dimensionality sweep for the reduced input configuration.

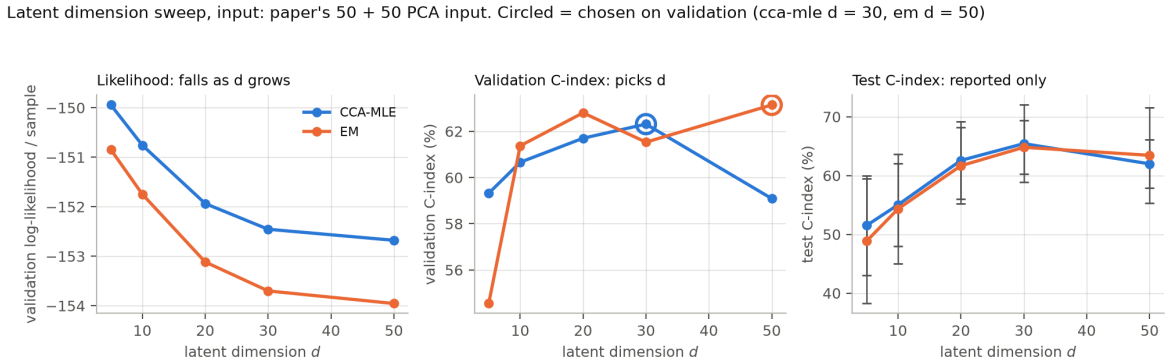


Figure 7: Latent dimensionality sweep for the paper PCA configuration.

The saved results select a latent dimension of 20 for CCA-MLE and 10 for EM in the reduced configuration. For the paper PCA configuration, the selected dimensions are 30 for CCA-MLE and 50 for EM.

The different selected dimensions show that the preferred representation size depends on the input configuration and parameter-learning method. A larger latent dimension does not necessarily produce better survival prediction, and the selected values should be understood as the outcomes of the current validation procedure rather than universal optima.

7.9 Posterior Entropy

Posterior entropy is included as an additional measure for examining uncertainty in the inferred latent representation.

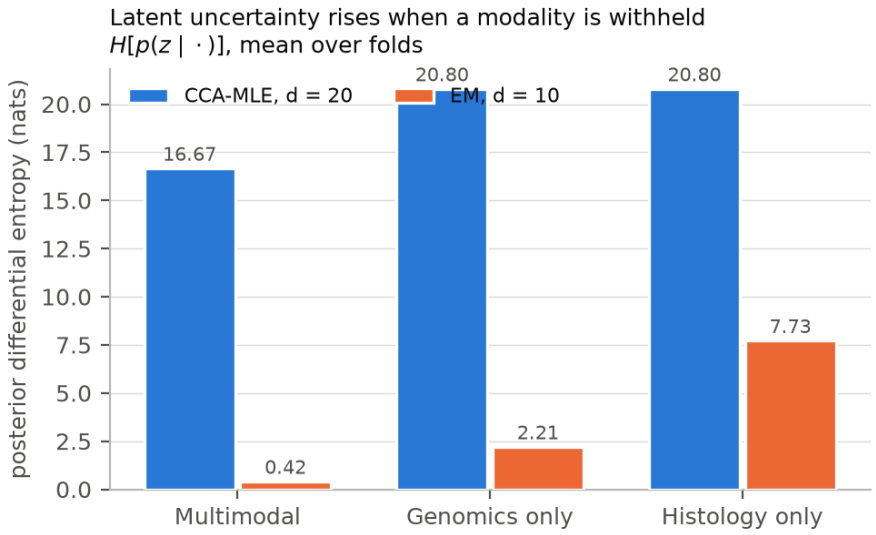


Figure 8: Saved posterior entropy analysis for the latent-variable inference settings.

The posterior entropy analysis provides a view of the uncertainty associated with latent-variable inference. Joint-modality and modality-specific inference use different sets of observations, and the resulting posterior distributions may therefore differ in their uncertainty.

Entropy should not be interpreted as a direct measure of survival prediction accuracy. A posterior with lower entropy is more concentrated under the assumed distribution, but that concentration alone does not establish that its latent representation predicts survival more accurately.

The figure is therefore used as a complementary analysis of inference behavior rather than as evidence of improved predictive performance.

7.10 Summary of Initial Findings

The initial results support the following observations:

- The implemented probabilistic models provide a functioning experimental framework for multimodal and modality-specific survival prediction.
- The non-PGM baseline achieves a higher mean multimodal C-index than both implemented PGM methods in the paper PCA configuration.
- In the reduced configuration, the EM-based model outperforms CCA-MLE in the multimodal setting, but the non-PGM baseline still achieves the highest mean multimodal C-index.
- The highest mean C-index among the listed methods occurs in the genomics-only EM setting for the reduced configuration, at 72.11%. This does not establish that genomics-only modeling is generally preferable.
- EM convergence differs substantially between the two input configurations, with all five PCA-configured folds converging and only two reduced-input folds converging.

- The selected latent dimensionality varies with both the input configuration and parameter-learning method.
- The paper-baseline reproduction agrees with the authors' stored per-fold evaluation results, while the original MATLAB training process was not independently rerun.

These findings establish an initial quantitative assessment of the M2 implementation. They also identify areas for further work, particularly EM convergence, the relationship between the learned representation and survival prediction, and evaluation using the full feature representation.

8 Changes Since M1 and Initial KPI Assessment

8.1 Transition from M1 to M2

Milestone 1 (M1) established the project problem, research direction, and proposed approach for multimodal breast cancer survival prediction using a Probabilistic Graphical Model (PGM). The primary goal was to investigate whether genomic and histology-derived features could be represented jointly through a shared latent variable and used for survival prediction.

Milestone 2 (M2) focused on translating this proposed approach into a technical implementation. The work progressed from defining the research problem to implementing the graphical model, developing parameter-learning procedures, performing latent-variable inference, and establishing an experimental pipeline for evaluating survival prediction.

The main change between the two milestones is therefore the progression from a proposed methodology to an implemented framework with initial quantitative results.

8.2 Progress Against the Original Project Goals

Table 10 summarizes the progress from the original project direction to the current M2 implementation.

The M2 implementation establishes the main technical components required for the planned experiments. However, implementation of these components does not imply that all planned extensions have been completed or that the PGM has demonstrated superior predictive performance.

8.3 Initial Key Performance Indicators

The initial key performance indicators (KPIs) focus on predictive performance, parameter-learning behavior, and the ability to reproduce the available reference-paper evaluation.

Table 11 summarizes the principal quantitative findings from the saved experimental results.

The results demonstrate that the project has progressed beyond model formulation to quantitative evaluation. At the same time, the predictive KPI remains an area for improvement

Table 10: Progress from M1 to M2

Project Component	**M1 Direction**	**M2 Status**
Problem formulation	Define multimodal breast cancer survival prediction as the project problem.	Retained as the central research problem.
Graphical model	Propose a shared latent-variable model connecting genomics and histology.	Implemented the model structure and its probabilistic formulation.
Parameter learning	Identify suitable parameter-learning methods.	Implemented maximum-likelihood estimation and EM-based learning.
Latent-variable inference	Investigate inference over the shared latent representation.	Implemented joint-modality and modality-specific inference procedures.
Survival prediction	Evaluate the learned representation for survival prediction.	Integrated the representations with a Cox proportional hazards model.
Experimental evaluation	Plan comparisons with the reference framework and alternative methods.	Obtained saved C-index results for PGM methods, modality-specific settings, and a non-PGM baseline.
Latent dimensionality	Determine a suitable representation size.	Evaluated candidate dimensions using validation-set C-index.
Full feature representation	Consider the use of higher-dimensional features.	Deferred fitting the full feature representation to M3 because of covariance-estimation and sample-size constraints.

Table 11: Initial M2 KPI assessment

KPI	**Observed Result**	**Assessment**
Multimodal C-index, paper PCA	EM: $63.44 \pm 8.09\%$; non-PGM baseline: $66.08 \pm 6.97\%$.	The EM-based PGM does not outperform the non-PGM baseline in this configuration.
Multimodal C-index, reduced input	EM: $67.58 \pm 7.76\%$; non-PGM baseline: $69.78 \pm 6.32\%$.	The EM-based PGM improves over CCA-MLE but remains below the non-PGM baseline.
EM convergence, paper PCA	Five out of five folds converged.	The recorded runs satisfy the configured convergence criterion.
EM convergence, reduced input	Two out of five folds converged.	Convergence requires further investigation for this configuration.
Reference-baseline reproduction	Maximum difference of 0.00 percentage points against the authors' stored per-fold C-index results.	The saved evaluation agrees with the available author-provided results. The original MATLAB training procedure was not rerun.
Latent dimension selection	Selected dimensions range from 10 to 50, depending on method and input configuration.	The preferred dimension depends on the model and input representation.

because the non-PGM baseline achieves a higher mean multimodal C-index in both input configurations.

The convergence KPI also reveals a difference between configurations. While all five paper PCA EM runs converged, only two reduced-input runs converged within the configured iteration limit. This finding motivates further investigation of parameter estimation and convergence behavior.

8.4 Changes in Experimental Scope

The experiments use two feature configurations: the paper PCA representation and a reduced representation containing genomic principal components and raw CellProfiler histology features. This allows the model to be evaluated under different input representations while retaining a common survival-prediction objective.

The full feature representation, consisting of 1,000 genomic features and 1,075 histology-derived features, has not been fitted using the current model during M2. Its evaluation has been deferred to M3 because of the dimensionality and covariance-estimation constraints documented in the project.

The current milestone therefore prioritizes establishing a functioning experimental pipeline and obtaining interpretable initial results over expanding the number of experimental settings.

8.5 Changes to Problem Framing and Research Positioning

The central problem remains multimodal breast cancer survival prediction using a shared latent-variable representation. The M2 findings do not require a change to this problem statement, but they refine how the implementation should be positioned.

In particular, the initial results do not establish that integrating both modalities always improves predictive performance. The results instead indicate that performance depends on the input configuration, inference setting, and parameter-learning method.

Consequently, the project is positioned as an implementation and evaluation of an existing probabilistic modeling framework, with further investigation of its behavior and potential improvements. Claims of superiority over the reference methods or the non-PGM baseline would require additional supporting experiments.

8.6 Overall M2 Assessment

M2 establishes the core technical foundation for the project: a specified graphical model, parameter-learning procedures, latent-variable inference, a survival-prediction pipeline, and saved quantitative and visual evaluation outputs.

The principal achievements are the implementation of the modeling workflow, the evaluation of multiple input configurations, and the reproduction of the available author-provided baseline evaluation. The principal unresolved issues are the lower multimodal C-index relative to the non-PGM baseline and the convergence difficulties observed in the reduced-input EM experiments.

These findings provide a concrete basis for planning M3. The next stage should focus on improving the reliability of parameter learning, examining the causes of convergence difficulties, and extending the experimental evaluation where technically feasible.

9 Limitations and Future Work

9.1 Limitations

Although Milestone 2 establishes the main technical foundation of the project, several limitations remain and must be addressed in the next milestone.

1. **Limited feature representation:** The current experiments use reduced representations of the genomic and histology modalities. The paper-based configuration uses 50 principal components from each modality, while the reduced configuration uses 50 genomic principal components and 215 raw CellProfiler histology features. Experiments using the complete feature representations of 1,000 genomic features and 1,075 histology features have not yet been completed. Their evaluation requires careful consideration of covariance estimation and the available sample size.

2. **Performance relative to the baseline:** The multimodal PGM has not consistently outperformed the non-PGM baseline in the current experiments. For the paper-based configuration, the EM model achieved a mean C-index of 63.44, compared with 66.08 for the non-PGM baseline. For the reduced configuration, the corresponding values were 67.58 and 69.78. Therefore, the current results do not establish a predictive advantage for the PGM.

3. **EM convergence:** The EM algorithm converged in all five folds for the paper-based configuration, requiring between 75 and 104 iterations. In the reduced configuration, only two of the five folds converged, while the remaining three reached the maximum iteration limit of 300 without satisfying the convergence criterion. This difference indicates that convergence and numerical stability require further investigation.

4. **Limitations of reference-paper reproduction:** The reference-paper comparison uses the authors' stored trained projection matrices, outputs, and survival-analysis code. The original MATLAB model-training procedure was not rerun. Consequently, the comparison reproduces the available reference artifacts rather than independently reproducing every stage of the original training pipeline.

5. **Modeling assumptions:** The graphical model assumes that genomic and histology features are conditionally independent given the shared latent variable. It also uses a Gaussian probabilistic CCA formulation. These assumptions provide a tractable modeling framework, but the current experiments do not establish that they fully represent the underlying biological relationships.

6. **Evaluation and generalizability:** The current evaluation uses five-fold results summarized by their mean and standard deviation. No statistical significance testing has been reported. The C-index evaluates the ranking of predicted survival risks, but it does not

by itself establish probability calibration or clinical usefulness. Furthermore, the experiments use the TCGA-BRCA cohort, so performance on independent patient populations remains unverified.

7. **Interpretation of posterior entropy:** Posterior differential entropy is included as a measure of uncertainty in the inferred latent representation. However, entropy values alone do not demonstrate improved survival prediction or clinical reliability and must be interpreted alongside the predictive evaluation.

9.2 Future Work for Milestone 3

The next milestone will focus on improving the reliability of the implementation, investigating the current results, and extending the evaluation where feasible.

1. **Investigate EM convergence:** Examine the folds that did not converge in the reduced configuration. Potential investigations include initialization, covariance conditioning, convergence criteria, and numerical stability. Any changes will be evaluated against the existing implementation to determine whether they improve convergence without compromising predictive performance.

2. **Evaluate the complete feature representations:** Extend the experiments beyond the current reduced inputs. Since the complete feature representations are high-dimensional relative to the available sample size, this step will require careful consideration of dimensionality reduction and appropriate regularization.

3. **Conduct systematic sensitivity and ablation experiments:** Evaluate how latent dimensionality and modality selection affect model performance. Compare multimodal predictions with genomic-only and histology-only predictions under consistent experimental conditions. Missing-modality evaluation may also be considered as an extension if the implementation and available data support it.

4. **Strengthen baseline comparisons:** Compare the PGM and non-PGM approaches using consistent data splits, preprocessing, and evaluation procedures wherever possible. Report the variability of results and use appropriate statistical analyses before drawing conclusions about differences in predictive performance.

5. **Improve reproducibility:** Document the experimental configuration, selected latent dimensions, convergence behavior, and saved outputs. Ensure that the experiment scripts and result-generation procedures can be followed consistently from the project repository.

6. **Assess generalizability where feasible:** If a suitable independent dataset becomes available, evaluate whether the learned representation and survival predictions generalize beyond the current TCGA-BRCA cohort.

9.3 Priorities for the Next Milestone

Table 12 summarizes the principal priorities for Milestone 3.

Table 12: Planned priorities for Milestone 3.

Priority	Task	Expected Outcome
1	Investigate EM convergence	More reliable fitting and a documented explanation of convergence behavior.
2	Extend feature representation experiments	An assessment of whether richer input representations are feasible and beneficial.
3	Perform sensitivity and ablation experiments	Better understanding of the contribution of each modality and the effect of latent dimensionality.
4	Strengthen baseline evaluation	A more consistent and transparent comparison of predictive performance and uncertainty.
5	Improve reproducibility and documentation	A clearer, repeatable experimental pipeline with traceable results.

9.4 Expected Direction

The main objective for Milestone 3 is not to assume that the PGM will outperform the baseline, but to determine under which conditions the model is reliable and whether its latent representation provides a useful basis for survival prediction. The next stage will prioritize convergence, fair evaluation, and careful interpretation of results before making stronger claims about the effectiveness of multimodal probabilistic graphical modeling for breast cancer survival prediction.

10 Conclusion

This project investigates the use of a multimodal Probabilistic Graphical Model (PGM) for breast cancer survival prediction using genomic and histology-derived features from the TCGA-BRCA dataset. The central idea is to represent the relationship between these two modalities through a shared latent variable and use the resulting representation for survival-risk prediction.

During Milestone 2, the project established a technical foundation for this approach. The graphical model structure was specified, Gaussian probabilistic CCA was implemented using maximum-likelihood and Expectation-Maximization (EM) approaches, and posterior inference was incorporated to examine the shared latent representation. A Cox proportional hazards model was used to evaluate survival-risk predictions. The experimental pipeline also supports comparisons across modalities, latent dimensions, and baseline methods.

The initial results show that the PGM can be evaluated using the established pipeline, but its predictive advantage has not yet been demonstrated. For the paper-based input configuration, the multimodal EM model achieved a mean C-index of 63.44, compared with 66.08 for the non-PGM baseline. For the reduced input configuration, the corresponding values were 67.58 and 69.78. These results indicate that the non-PGM baseline performed better in both reported comparisons. Therefore, the current findings do not justify claiming that the PGM provides superior survival prediction.

The experiments also revealed an important implementation challenge. Although EM converged in all five folds for the paper-based configuration, convergence was less reliable for the

reduced configuration. This finding highlights the need to investigate numerical stability, convergence behavior, and the effect of input representation before drawing stronger conclusions about the model.

Overall, Milestone 2 has moved the project from the proposed modeling approach toward an implemented and experimentally evaluated framework. The next milestone will focus on improving the reliability of EM, extending the feature-representation experiments where feasible, and strengthening the evaluation against baseline methods. These steps will help determine whether the shared latent representation offers a useful contribution to multimodal breast cancer survival prediction.

11 References and Repository Links

References

[1] Reference research paper on multimodal genomic and histology-based breast cancer survival prediction.

[2] National Cancer Institute, *The Cancer Genome Atlas (TCGA)*. Available at: Official TCGA website.

[3] Group G02, *CSE-516 PGM Healthcare Breast Cancer Project*. Source code, documentation, and experimental results. Available at: Group GitHub Repository.

11.1 Project Repository and Resources

The following resources provide access to the project documentation, implementation, and saved experimental results.

- **Complete Group Repository:** Open the GitHub Repository
- **Milestone 1 Documentation:** View M1 Files
- **Milestone 2 Documentation:** View M2 Files
- **M2 Source Code:** View Implementation
- **M2 Experimental Results and Figures:** View Results
- **Processed Dataset Documentation:** View Dataset Documentation